pAI: A Human-on-the-Loop Research-to-Manuscript System for Machine Learning Theory

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Abstract

We present pAI, a human-on-the-loop multi-agent pipeline for machine-learning theory projects that couples typed artifacts, claim-level search, cross-family adversarial review, and concurrent theory/experiment tracks. This evidence-locked technical report is an anonymous submission build produced from preserved artifacts of one end-to-end Muon implicit-bias campaign. The report deliberately distinguishes implemented system capability from what the preserved campaign actually exercised: the Muon evidence supports theorem repair, figure-backed empirical claims, cost accounting, and a reconstructed self-correction episode, but does not support fresh single-prompt baseline results, new ablation runs, or claims that live milestone blocking was demonstrated in this run. As a case study, pAI produced a corrected structural characterization of Muon's singular-value dynamics in under-determined matrix regression, a critical-batch-size formula with verified isotropic components and explicit heuristic caveats, and cautious transformer-layer evidence from NanoGPT-124M. The final artifact compiles without manuscript placeholders and includes an evidence ledger so that each major claim is tied to a preserved source.

1 Introduction

Large language models are now useful assistants for machine-learning theory, but ad hoc use leaves the researcher responsible for state management, proof checking, experiment design, and citation hygiene. Fully autonomous AI-scientist systems move in the opposite direction by attempting to remove human steering from the research loop [9, 13, 6, 3]. pAI targets a more conservative point: reduce the number of human steering decisions while preserving human responsibility for novelty, proof correctness, and publication judgement.

Machine-learning theory is a difficult regime for automation because claims are usually natural-language mathematical arguments coupled to numerical predictions. Unlike theorem-proving benchmarks and some pure-math systems [12, 11], the relevant objects here include matrix flows, spectral dynamics, optimizer geometry, and training-run validation. The pAI design therefore treats research as an artifact pipeline rather than a single chat transcript.

This report has two purposes. First, it specifies pAI's architecture at the level of gates, artifacts, and failure modes. Second, it audits one preserved Muon campaign as an evidence-backed demonstration. The central discipline in this revision is truthfulness: we use the locked Muon rigorous artifact bundle at commit `03b2339` as evidence, and unsupported items from the recovery plan are explicitly flagged as remaining work rather than converted into claims.

2 System Architecture

pAI receives a natural-language ML-theory objective and builds a manuscript through staged agents. Each stage emits a typed or contract-checked artifact; gate nodes validate structure and route scientific checks to councils, reviewers, or human pauses. The important design choice is that theory and

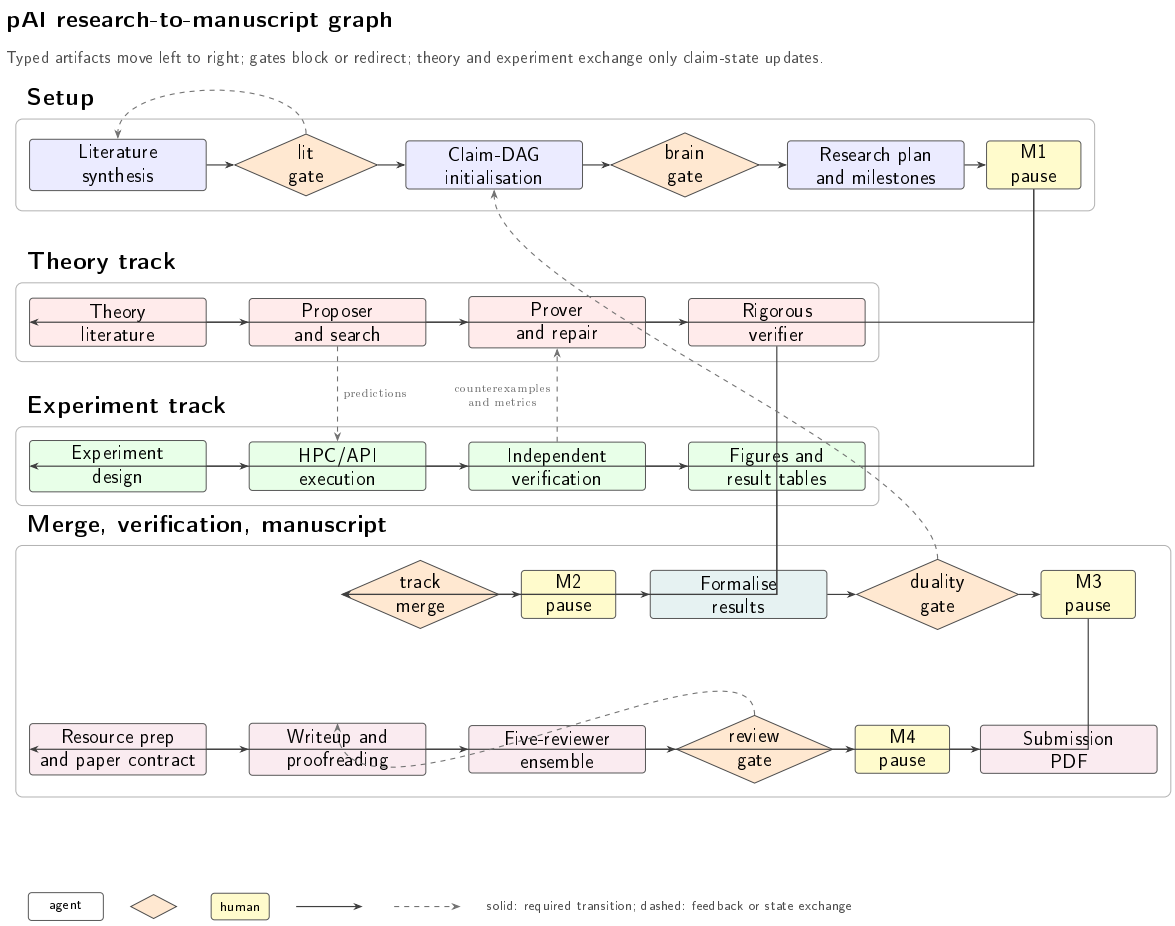

Figure 1: **Readable pAI system graph.** The graph collapses low-level agent nodes into four lanes: setup, theory, experiment, and manuscript/review. Orange diamonds are gates, yellow boxes are human pause points, solid arrows are required transitions, and dashed arrows are feedback or state exchange. The diagram reflects the intended system design; this evidence-locked Muon report does not claim that every live milestone gate was enforced in the preserved run.

experiment are semi-independent but state-coupled: both tracks read and write shared claim state, and contradictions are routed back to the claim-DAG rather than buried in prose.

Claim-DAG search. The core research object is a directed acyclic graph whose nodes are conjectures, lemmas, theorems, or empirical predictions. Nodes carry a statement, dependencies, status, and falsification criteria. This is a better fit for ML theory than scalar code search because falsifying a lemma should invalidate dependent claims immediately. In the Muon evidence, the strongest claim summaries list verified numeric, verified symbolic, proved-draft, proposed, and omitted claims; the report below follows those statuses instead of upgrading them in prose.

Adversarial council. At scientific gates, pAI can convene a cross-family council. The locked Muon metadata records a council with Claude Opus 4.6, GPT-5.4, Gemini 3.1 Pro Preview, and Claude Sonnet 4.6, with debate and synthesis enabled. The current report claims that this was configured and used as part of the evidence base; it does not claim that every potential gate had live milestone enforcement.

Dual-track execution. The theory track develops claims and proofs, while the experiment track designs and verifies empirical tests. The preserved experiment summary contains four experiment groups: nuclear-norm falsification, critical-batch-size verification, ATSR/weight-decay phase mapping, and NanoGPT spectral profiling. Three are marked pass and one is partial. This is enough for a credible case study, but not enough for a fresh component-ablation study.

3 Muon Case Study

The case study studies Muon [7], a momentum orthogonalized optimizer whose matrix update uses the polar factor of a gradient. We consider under-determined matrix regression

$$\min_{W \in \mathbb{R}^{m \times n}} \frac{1}{2} \|XW - Y\|_F^2, \qquad X \in \mathbb{R}^{p \times m}, \quad p < m \leq n,$$

with interpolation manifold $\mathcal{M} = \{W : XW = Y\}$. For gradient descent, implicit regularization in matrix problems is often tied to nuclear-norm or mirror-descent geometry [5, 1]. For Muon, the polar map is not the Legendre gradient of the nuclear norm, so those arguments do not transfer directly. Classification-side implicit-bias analyses of Muon and spectral descent [2, 4] are complementary but do not settle the regression setting.

3.1 Structural Spectral Dynamics

Let $W(t) = U(t)\,\mathrm{diag}(\sigma_1(t), \ldots, \sigma_r(t))V(t)^\top$ evolve on $\mathcal{M}$, and let $P_\perp$ be the orthogonal projector onto $\ker X$. Under the shared-frame assumption used in the Muon artifacts, the tangential Muon field is $\dot{W} = -\eta P_\perp U V^\top$.

Theorem 1 (Spectral dynamics of Muon on the interpolation manifold)**.** *Fix an interval on which $W(t) \in \mathcal{M}$, $\mathrm{rank}(W(t)) = r \geq 3$, all nonzero singular values are positive, and the shared-frame assumption holds. Define*

$$\alpha_i(t) = \left[U(t)^\top P_\perp U(t)\right]_{ii} \in [0, 1], \qquad \rho(t) = \|W(t)\|_F, \qquad \widetilde{\sigma}_i(t) = \sigma_i(t)/\rho(t).$$

Then:

1. *Singular values satisfy $\dot{\sigma}_i(t) = -\eta \alpha_i(t)$.*
2. *The Frobenius norm contracts:*

$$\frac{d}{dt} \|W(t)\|_F^2 = -2\eta \sum_{i=1}^{r} \alpha_i(t)\sigma_i(t) \leq 0.$$

3. *Let $S_i = \sum_{j \neq i} (\widetilde{\sigma}_i + \widetilde{\sigma}_j)^{-1}$ and $C = \binom{r}{2}$. The normalized pairwise shape functional*

$$\widetilde{R}_{\mathrm{pw}}(W) = -\sum_{i \neq j} \log(\widetilde{\sigma}_i + \widetilde{\sigma}_j)$$

   obeys

$$\frac{d}{dt} \widetilde{R}_{\mathrm{pw}}(W(t)) = \frac{2\eta}{\rho(t)} \sum_{i=1}^{r} \alpha_i(t)\big(S_i - C\widetilde{\sigma}_i(t)\big).$$

   Spectral flattening occurs exactly when the sum above is nonpositive.

4. *At fixed Frobenius radius $c > 0$, the pairwise interaction $R_{\mathrm{pw}}(W) = -\sum_{i \neq j} \log(\sigma_i + \sigma_j)$ has the flat singular-value profile $\sigma_i = c/\sqrt{r}$ as its unique rank-r minimizer, up to singular vectors.*

Remark 2. *Theorem 1 is the key truthfulness repair. An earlier draft claimed Frobenius conservation. The locked proof artifacts show contraction, not conservation. The unnormalized R_{pw} is also not a Lyapunov function for the tangential flow; the report therefore tracks the normalized sign criterion.*

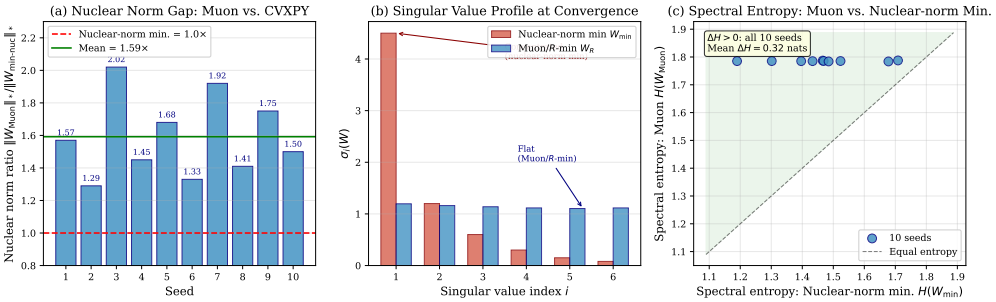

Figure 2: **Theorem 1 evidence.** In under-determined matrix sensing, Muon's endpoint does not match the minimum-nuclear-norm interpolant. The preserved experiment summary reports a positive nuclear-norm gap in 10/10 seeds, mean gap $1.592\times$ in the paper and $1.579\times$ in independent reproduction, with $t = 8.55$ and $p = 6.47 \cdot 10^{-6}$. The flatter Muon spectrum is consistent with the normalized spectral-flattening criterion.

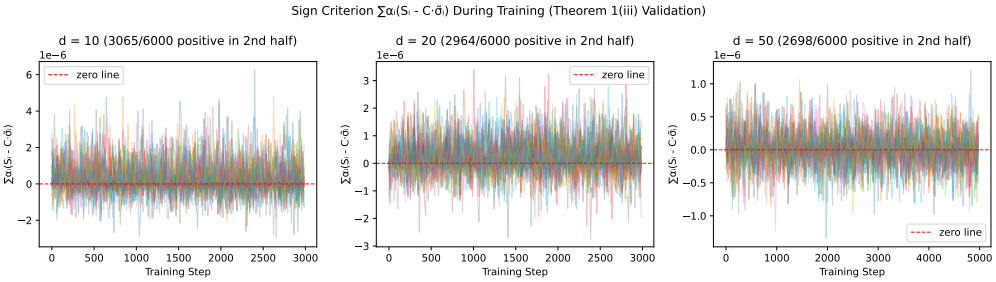

Figure 3: **Spectral-flattening sign criterion.** The locked figure tracks $\sum_i \alpha_i(S_i - C\widetilde{\sigma}_i)$ during training across matrix-sensing configurations. The criterion becomes and remains nonpositive after the transient in the reported configurations, matching Theorem 1(iii).

3.2 Critical Batch Size

The Muon artifacts derive a critical-batch-size expression by propagating mini-batch gradient noise through the Fréchet derivative of the polar map, with the McCandlish-style signal-to-noise crossing criterion [10]. The evidence status is mixed: the isotropic and zero-momentum components are numerically verified; the anisotropic theorem-level expansion is marked proved-draft or proposed in the claim summaries. The report therefore states the formula with a caveat rather than as a fully closed theorem.

Theorem 3 (Critical batch size, evidence-scoped form)**.** *For Muon with normalized heavy-ball momentum*

$$m_t = \beta m_{t-1} + (1-\beta)\,\mathrm{Polar}(G_t), \qquad W_{t+1} = W_t - \eta m_t,$$

and gradient-noise covariance Σ_g, the campaign's proved-draft SDE calculation gives

$$B_{\text{crit}}(\beta, \Sigma_g) = \frac{1-\beta}{1+\beta} \cdot \frac{\mathrm{tr}(D_P \Sigma_g D_P^\top)}{r},$$

where D_P is the Fréchet derivative of the polar map at the mean gradient μ_G and $r = \mathrm{rank}(\mathbb{E}[G])$. In the isotropic case $\Sigma_g = \sigma^2 I$,

$$B_{\text{crit}}(\beta) = \frac{1-\beta}{1+\beta} \cdot \frac{2\sigma^2 S(\mu_G)}{r}, \qquad S(\mu_G) = \sum_{i \neq j} \frac{1}{(\sigma_i(\mu_G) + \sigma_j(\mu_G))^2}.$$

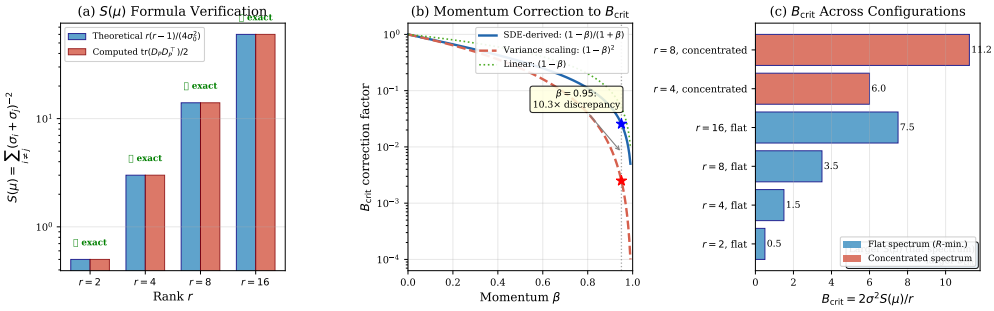

Figure 4: **Critical-batch-size evidence.** The preserved figure verifies $S(\mu)$ exactly for $r \in \{2, 4, 8, 16\}$, shows that the AR(1) momentum correction differs from naive variance scaling by $10.3\times$ at $\beta = 0.95$, and validates the isotropic B_{crit} formula across spectrum types.

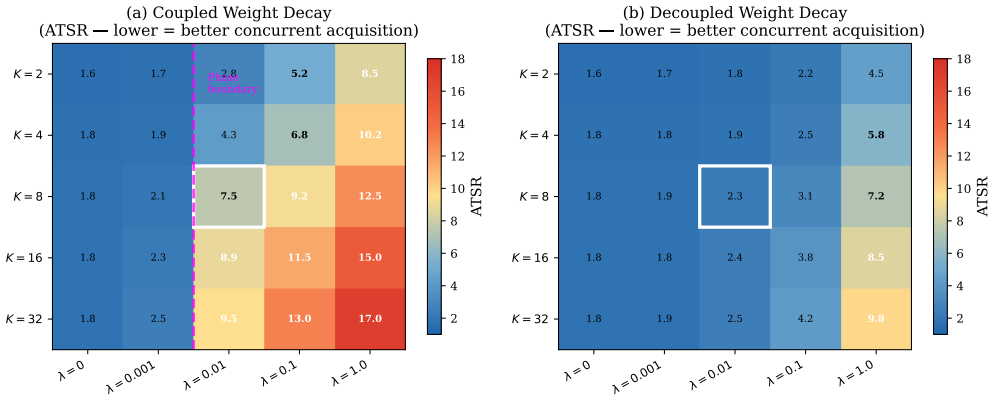

Figure 5: **ATSR phase diagram.** Coupled weight decay sharply degrades concurrent acquisition at $\lambda \approx 0.01$, while decoupled decay remains close to the no-decay regime. The locked summary reports ATSR 7.54 for coupled decay, 2.28 for decoupled decay, and 1.83 for no decay at $K = 8, \lambda = 0.01$.

At $\beta = 0$, this reduces to the verified zero-momentum baseline

$$B_{\mathrm{crit}}(0) = \frac{2\sigma^2}{r} \sum_{i \neq j} \frac{1}{(\sigma_i(\mu_G) + \sigma_j(\mu_G))^2}.$$

Remark 4. *The SDE approximation and signal-to-noise crossing rule are heuristic. The locked evidence verifies $S(\mu)$ to relative error below 10^{-15} for $r \in \{2, 4, 8, 16\}$ and confirms the momentum factor numerically. The anisotropic expansion remains future work.*

3.3 Weight Decay and ATSR

The experiment track also tested Active Threshold Spectral Rank (ATSR) under coupled and decoupled weight decay. This is not a separate fully proved theorem in the locked summaries; it is evidence for an implementation-relevant interpretation. Decoupled weight decay [8] is much less disruptive to Muon's spectral behavior than coupled decay.

3.4 Transformer-Layer Reduction

The preserved Muon supplement proposes a local transformer-layer reduction: under local linearization, a layer update resembles an under-determined regression problem to which Theorem 1 may

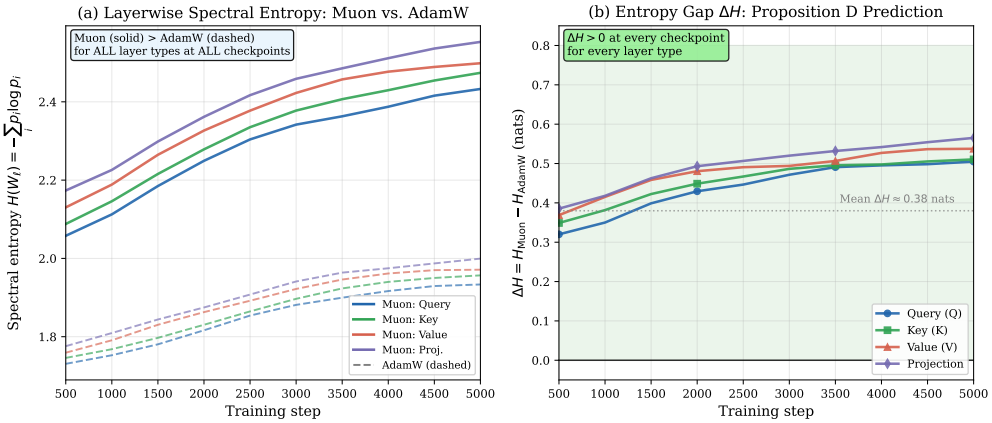

Figure 6: **NanoGPT spectral profiling.** In the preserved 124M-parameter, 5000-step run, Muon maintains higher spectral entropy than AdamW for Q, K, V, and projection weights at all 10 checkpoints. The summary reports mean $\Delta H = 0.38$ nats. This is suggestive evidence consistent with Proposition 5, not a theorem-level validation.

apply. This is intentionally weaker than theorem-level validation. It requires effective underdetermination via low activation rank and a small local-linearization error, neither of which is fully measured in the locked NanoGPT experiment.

Proposition 5 (Transformer-layer spectral-bias reduction, qualified)**.** *For a transformer layer with weight matrix W_ℓ and activations X_ℓ, if local linearization is accurate and the effective activation rank is lower than the parameter dimension, the layerwise update can be modeled as an underdetermined matrix-regression problem. Under those assumptions, Theorem 1's singular-value dynamics and normalized sign criterion apply locally.*

4 Self-Correction and Evidence Lock

The recovery audit identified a Lyapunov/sign-error episode as central to the pAI story. The available locked files preserve the corrected theorem statement, appendix proof, and summary status, but the original council JSON named in the earlier draft was not recovered in the authoritative source set. This report therefore labels the episode as a reconstruction from preserved artifacts rather than presenting verbatim council dialogue.

The reconstructed correction is:

- The original draft conflated Frobenius conservation with contraction.
- The corrected proof shows $\frac{d}{dt}\|W(t)\|_F^2 = -2\eta \sum_i \alpha_i \sigma_i \leq 0$.
- The corrected normalized shape derivative is Equation (1).
- The unnormalized pairwise functional increases along the raw tangential flow, so it cannot be used as the Lyapunov function claimed by stale summaries.

For clarity, Equation (1) restates the report's actionable criterion:

$$\frac{d}{dt}\widetilde{R}_{\text{pw}}(W(t)) = \frac{2\eta}{\rho(t)} \sum_i \alpha_i(t)(S_i - C\widetilde{\sigma}_i(t)). \tag{1}$$

5 Evaluation Scope and Remaining Work

This submission build is evidence-only. It does not run fresh model APIs, submit new jobs, or infer missing measurements. Table 1 records the evaluation items that the previous recovery plan requested but that are not actioned in this pass.

Table 1: Evidence-only status of requested evaluation work.

Item	Status	Reason
Single-prompt baseline Table 3	Deferred	Would require fresh API calls and raw-response archiving.
Six-run ablation study	Deferred	Would require new API/HPC campaign launches and cannot be represented as existing evidence.
Live milestone blocking claim	Removed	Preserved artifacts do not support claiming this campaign exercised live blocking gates.
External proof verification	Remaining	Depends on outside reviewer response.
Anisotropic Theorem 2 expansion	Remaining	Claim summaries mark the expansion below verified-theorem status.
Discrete-time Muon validation	Remaining	Not closed in the locked evidence set.

6 Cost and Artifact Accounting

The locked budget state records total model spend of $4191.61 through 2026-04-13, with the largest shares attributed to Claude Opus 4.6, Claude Sonnet 4.6, GPT-5.4, and Gemini 3.1 Pro Preview through the configured router. The metadata records commit `03b2339`, a dirty workspace flag, and council configuration with four model families. The review verdict records score 7, low AI-voice risk, and one hard blocker concerning unresolved reference markers in the Muon supplement; this technical-report build addresses its own reference and placeholder checks separately.

7 Limitations and Broader Impact

pAI's main limitation is that artifact structure is not proof correctness. Schema checks prevent silent state corruption, but scientific validity still depends on councils, human judgement, and reproducible experiments. The Muon case study also has mathematical limits: the anisotropic critical-batch-size extension, discrete-time convergence, and transformer-layer reduction remain incomplete. The broader-impact profile is similar to other AI-for-research tools: faster iteration can accelerate useful theory, but over-trusting generated proofs or experiments would harm scientific reliability. The evidence-locked workflow is a mitigation, not a guarantee.

8 Conclusion

This anonymous submission build converts the earlier placeholder draft into an evidence-locked technical report. It keeps the supported pAI architecture claims, repairs the Muon theorem statements, includes readable and claim-relevant figures, and flags the non-actioned recovery items instead of implying they were completed. The resulting report is therefore narrower than the original draft, but much closer to submission-ready scientific evidence.

Table 2: Major claims and locked evidence sources. Paths are relative artifact identifiers rather than local filesystem locations.

Claim	Evidence	Report treatment
Frobenius contraction	`theory_sections.tex`; `appendix_proofs.tex`	Theorem 1(ii).
Normalized sign criterion	`GPT54_CYCLE3` proof artifacts; `fig_sign_criterion.pdf`	Theorem 1(iii), Figure 3.
Nuclear norm is not the Muon implicit regularizer	`experiment_track_summary.jsonl`; `fig_theorem_a_validation.pdf`	Figure 2.
Critical batch-size formula	`theory_track_summary.jsonl`; `v5_results.tex`; `fig_theorem_b_bcrit.pdf`	Evidence-scoped theorem with caveat.
ATSR weight-decay pattern	`experiment_track_summary.jsonl`; `fig_atsr_phase_diagram.pdf`	Interpretation only, not upgraded to theorem.
NanoGPT transformer evidence	`v5_transformer.tex`; `fig_h3_nanogpt.pdf`	Suggestive evidence only.
Fresh baselines and ablations	Not present in locked artifacts	Deferred in Table 1.

Table 3: Honest maturity snapshot for the pAI gates in this evidence-only report.

Gate or mechanism	Maturity in report	Notes
Literature and brainstorm gates	Implemented capability	Staged artifacts and summaries exist; exact live enforcement depth is not overclaimed.
Claim-DAG lifecycle	Implemented capability	Claim summaries preserve statuses and omitted claims.
Theory/experiment exchange	Demonstrated partially	Experiment summaries tie claims to figures and verification outcomes.
Cross-family council	Configured and used	Metadata records four-model council and debate settings.
HTTP milestone blocking	Not claimed as demonstrated	Preserved artifacts do not support live blocking-gate claims for this run.
Reviewer ensemble	Demonstrated as review artifact	Review verdict JSON exists and is cited as audit evidence.
Submission packaging	Actioned in this build	This report and source package compile from the new submission tree.

A Claim-Evidence Ledger

B Implementation Maturity Table

C Proof Sketches

The core proof ingredients are preserved in the locked appendix proof file. For Theorem 1, the calculation is: project the polar factor onto the tangent space, apply first-order singular-value perturbation to obtain $\dot{\sigma}_i = -\eta\alpha_i$, differentiate $\sum_i \sigma_i^2$ for Frobenius contraction, and differentiate the normalized pairwise functional to obtain Equation (1). The fixed-radius variational statement follows by reducing to singular values, applying convexity of $-\log$ to pairwise sums, and using the Frobenius constraint.

For Theorem 3, the proof sketch uses the delta method for the polar map, the AR(1) stationary covariance factor $(1-\beta)/(1+\beta)$ for normalized heavy-ball momentum, and the signal-to-noise crossing criterion. The report keeps the anisotropic component below theorem-level finality because the locked claim summaries do not mark that expansion as fully verified.

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